ActInf Textbook Group ~ Cohort 9 ~ Session 1 (Chapter 1 and overall) ~ 8/21/2025

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Daniel, um, any suggestions on which resources you'd recommend like glancing through as like a starting point? Like if you I had some time before after reading the chapter, is there like a a good like first place to look because there's so much resource on here honestly. It's like >> Yeah, that is a good question. And I think it's more about sort of what what you want to get out of out of active inference or approaching. So you said you're a master's uh you're doing compsai MIT. Um do you I mean most people are not using mat lab so much anymore but there's a nice uh there was a nice and I'll share this in the chat um step-by-step tutorial on active inference put out by uh Fristen and and and some others including Ryan Smith. Uh, and Ryan Smith's a pretty um has a bit of a strong reputation in the community. Um, he does a lot with computational psychiatry, but along the way he's repeatedly given like tutorials at at Yale and at Zurich and so on uh on how to actually do this stuff because there is at times a bit of a gap between actually learning this versus like knowing how to apply it in in code etc. and getting into the models. So even if you don't want to take the mat lab route necessarily, it really breaks down a lot of sort of the core maths and then along the way you kind of get this commentary on like sort of use cases and things like that. So I think that's a nice intro. Um, on the more theoretical side, uh, active inference, a process theory, which is a, um, it's a bit of a long read, but it's uh, it's it's a core text that came out 5 years before this textbook, but um, people kind of refer back to it very often. And it's sort of itself a a compendium of of what Fristen and others uh had been working on up to that point and tries to synthesize it where I think this textbook becomes a much stronger kind of synthesis of the last five years after that plus that. But uh 50 pages is less than 200. So uh you know worth skimming around in there. Thanks. >> For sure. >> Cool. Um, all right. So, thanks again everyone for joining. I'll just I'll do I mean this is such a short initial chapter in this book. Uh we we're told during this chapter like a general rundown of what each chapter is going to kind of be about. But the main thing is to notice that the first half of the book being chapters 1 through 5 versus chapters 6 through 10 are quite different in certain respects in that the first half is intended to provide more of like a formalization of active inference. get a lot of the core uh ideas and the theory and the framework across and it does quite quickly by chapter two we start getting into the maths especially looking at uh baze uh bases rule or basis theorem and getting into kind of like a basian framework and thinking about uh also probability distributions but nonetheless it's still it's still very kind of prosaic relative to the second half of the textbook which is where uh it it much more strongly discusses particular models uh mainly exemplar models that are used in active inference. It refers to uh published research up to that point to kind of give uh more expansive explanations of these different models. Uh not just to get you familiar with what's already been done, but to kind of break down their architectures and and kind of cases of of what they were used to study uh in order to provide you with sort of fodder or information on how you might construct your own model. even if you're trying to do something more novel, uh there are many kind of components and and core concepts to be taken away from the second half of the book. Um but all of that being said, uh we'll I'll I'll talk about some of the core ideas that we're already introduced to in this first chapter. Um we know that that active inference is a normative framework. They they describe uh this sort of neats versus scruffy's debate. Uh I was not familiar with that kind of debate until I read this book, but I it it it clicked and and it's what I find very often in social sciences where people are much more contentious uh sort of with one another and there's so many different angles to to approach different phenomena. Um but in any case it's to uh it mainly approaches um concepts of perception and action which we're quite familiar with those who who study neuroscience especially perception uh sometimes action is kind of treated separately or or there are different kinds of models that might be used or or or different like uh augmentations of perception models to look at decision making. But but it can boil down for some to just perception. Um for active inference it's both. So active inference aims to um um kind of overcome this sort of classical or traditional notion of taking the brain or the organism or the entity as a passive observer of sensory information. uh rather there's perception in that sense but there's also action in that uh you know we as humans or or or or cellular organisms or entire cultures uh will will uh act in order to realize their goals or to seek information which are two very important kind of notions in that that more clearly uh are comprehensible whenever you start to get the the maths of how they break that down but for the time being it's to recognize we have perception and action and then with action we have this kind of subdivision into information seeking and and goal directed behavior which work together and and can quote unquote naturally balance one another. So um for those who are familiar with sort of um kind of the machine learning reinforcement learning paradigms like with reinforcement learning there's frequently uh you know there there will be kind of these rules applied to say what what does it mean to have exploratory behavior versus goal directed behavior. Um, free energy minimization involves kind of like a so-called natural way of of balancing that or determining what those things are as opposed to forcing some kind of stochastic rule that says like, oh, 5% of the time do something random so you don't get caught in a a local minima. Um, >> can I ask a question about that? Actually, I I had that on my notes. Um, >> yeah. Yeah, go for it. I I also have feel like I've had that dissatisfaction with reinforcement learning about like if it feeling like a hacky way of of trying to put in exploration. Um but then so later on the chapter um I think there was something very similar to what you said about essentially that that balance of exploration exploitation like in some sense naturally falls out of this um when you when you are thinking about planning. Um but then with the way the chapter opens and also the way you just described it, it kind of sounded to me like it was sort of just like declared um that that those are two the two goals that are being pursued by action. Like on the one hand it's like you know getting sense data that corresponds with goals, the other hand getting information which feels a little bit like like building it in by design. Not that that's not that there's anything like inherently wrong with that, but I guess I was um wasn't didn't feel like I was clear on like is that something that like naturally falls out or is that something that's kind of built in the way? >> That's a I I see like like the the difference between like naturally falling out versus like being built in. So I would it might be a little bit early to get to this given it's chapter one but um so this is uh this these are the various breakdowns of what's called expected free energy. Uh that and variational free energy are kind of the core quantities here. And the only reason why there there's a a division between those two, why we have two different free energy minimization rules um instead of one is that this is sort of expected free energy is sort of the free energy of the future uh which is a bit different and we there there are organisms who do not necessarily even have capacities like the kind of you know neurobiological capacities or otherwise to plan. um you know very cellular organisms like tend to lack this and they're a bit more reactive to their immediate environment. Uh but in any case for those who who do have that capacity um it can be broken down. We we'll ignore you know uh the vast majority of this equation but just for the starting line we have information gain and then we have pragmatic value and uh information gain would kind of be the equivalent of like seeking information. Uh whereas pragmatic value would be more in line with kind of how you stated the the goaldirectedness that is like bringing expected observations or observations you want to see uh you know in line with with reality or the other way around acting in order to realize what you want to see right so given that this is kind of scored over what policy to take what action to take what sequence of actions to take it's kind of like saying for each action you get you you get both of these, right? Um, so, so I guess we could say it's sort of built in in the sense of like, well, you know, if we define expected free energy this way, then this is one way that that can kind of be uh um the terms can kind of be assembled such that that this is what they look like and that's what they equate to. Um, I'm still working on my ability to kind of use like everyday examples of how to describe these things, right? So it's like information gain would be like let's start with pragmatic value. I'm hungry. I eat some food. Um great you know that that was pragmatic value by eating the food. Uh information gain. I'm hungry. Uh I need to find food. I need to locate where it is at. That's not immediately uh accomplishing my goal, but it is on the path there. It minimizes my uncertainty about the world around me and how to accomplish the goal. And so all of these different policies or actions uh are assumed to have their their be scored by expected free energy. Um and so both of those terms are there. So there's no like epsilon greedy like rule here going on. Uh it's it's like this is how the maths play out. And in terms of like code, it's like just infer policies like run a line and that's this is what you you end up with. Um a and like given that there's some here who have already read the textbook, some of this will already look familiar then. And so it's kind of like the this C matrix uh which COD is annoyingly kind of putting the pencil over, but the the C is something like preferences or like having goals. It's a it's a in this case for a discrete state space setting it's a distribution probability distribution over different potential observations you know uh being uh I I observe my stomach is growling versus it's not it's actually sad and feels good I'm I'm full uh you know higher probability for the latter because that's what I want like that's a very simplistic simplified example um and so the y tilda would be oh what do I expect do I expect that that uh by doing this action I will elicit a new observation that is I'm sated I'm good uh right so if if this is very high noting that this is a negative term then we are by by increasing the pragmatic value we're we're reducing this overall quantity right um with information gain of course looks a little bit more complex if good old um Colbeck Leebler divergence which is just kind of it's an equation that essentially just kind of measures the difference between two different distributions. Uh this first one is uh what do I think's going on in the world X conditioned on the actual observations sensory observations I received Y and pi the action I took versus just what do I think is going on in the world given uh the action I took. So it's kind of like we want to be able to do things and and and they link up with with the world. And so when we engage in action, we do things in order to change uh hidden states of the world. I my stomach growling is a sensory observation. My idea that I'm hungry or not hungry would be a hidden state in this case, right? There's no there's no sense I I have to relate things. So that's perception. Um but once I take in new data, what actually happened now? Right? So this is it's a form of updating one's beliefs between what what what is it that I think my actions do to change the world given the new data I received so that I can update things relative to just what do I think regardless of the sensory observation I'm receiving if I do something what how will the nature of the world change. So it's it's information gain is kind of just like taking actions to elicit observations the why uh and figure out did you learn anything new. If we increase that once again that's a negative term and so we end up reducing um expected free energy as well. So >> thanks Andrea looks like Muhammad has his hand up. >> Yeah. Yeah. Go for it. Um >> yeah I think I don't know like we maybe jumping quite ahead but I was just curious about since we've gotten to the reinforcement learning part here uh like does like the whole framework of active inference actually posit how do you even construct the state space in the first place like how do you even discretize your whole continuous space like since you're tracking probabilities like you can't track it onto everything you have to discretise your your whole world around you does it actually posit how you do that process or or we're just assuming okay given this state space how should the model update its probabilities over these possibilities. Um so yeah I'm just curious about if we have any ideas of in this framework that how do you even decide how do you construct that state space in the first place? >> Yeah know that that that is a great question. Um and so so first like and for those who like read the textbook or we'll keep going with it like we'll they'll notice that chapter 7 and chapter 8 are about models but in very different settings. Chapter 7 being about discrete state spaces and then chapter 8 being about continuous. And so it's like we we already like have this kind of division between how to look at a state space construction in the first place uh and and making a choice between the two uh with the continuous state spaces. That's where like the phrase predictive coding has come up frequently uh already during this this meeting and some folks have brought that up. that is going to be much more aligned with the notion of like continuous models versus discrete state spaces where a lot of these equations um describe both um there's another paper I I should probably actually um share with you all for anyone who's interested but there is a um and this is Ryan Smith again but there's a there's a nice paper that he put out last year um on the status of predictive coding meaning uh the empirical status of of predictive coding. And the whole sense of that is you know um unlike a kind of cultish uh you know take that this is more of a like well these models need to be fit to empirical data. They need to be validated. They should be benchmarked versus you know other standard models that are that are comparable but nonetheless you don't necessarily assume free energy minimization etc. Um, and it's he so he covers a lot of this in a really nice kind of succinct way. And uh um all that said um one free energy does kind of play out similarly in each of these models is just the equation for how you handle the state space. Um, but it's it's not what I mean to say is just because a model is continuous and another model is discreet, it doesn't necessarily mean that they are like intrinsically entirely different aside from how you're you're you're kind of dividing up the state space. Um then another kind of answer to the question in terms of what choices to and and another bit which is um the idea of having like hybrid models where you have like for example a common construction that comes up again way later in the textbook but um having like a kind of lower level model that takes in like s continuous sensory observations um you know just sensory receptors etc which tend to you know typically we empirically we we measure those or or even record those and more continuous spaces those might be received by a lower level continuous model but then through inference be linked up to a higher order discrete state space model. So you imagine like making like a trying to model a I don't know a mouse in a T-mas is a is a frequent example that's actually carried out in in laboratories and so on. And it's like, well, the mouse, you know, encounters a continuous world that it's within and continuous observations in it. Yet, it must make a discrete choice about should I go left or right in his teammates in order to get a discrete reward that is like, you know, some kind of cheese or or or or uh something sweet uh that's in one arm of the maze. So, th those kinds of structures are very much possible and employed. Um I've been playing around with them myself recently. Now and then finally another question is like what is the particular phenomena of interest that you have and what decision should you make in order to confront that using uh a generative model and making what kinds of uh architectural decisions to to actually approach that system of interest. Um so if you're you know if you're working with neuroiming data which tends to be viewed as continuous data even though there are like sampling rates and it some minute level actually it's quite discreet and could be represented as like know individual uh rows in a in a in a matrix or spreadsheet structure but but nonetheless we we tend to analyze it as continuous um you know so chapter six gets into a lot of that before it gets deep into the models it's sort recipe for active inference describing like well you do you do have to make various decisions right like you have to make various decisions on like what should the state's face look like and and and how how should we construct that what makes sense here relative to the literature that's already been published. So, so in that sense, it's not trying to claim that there's any kind of like magical, you know, oneandone approach to modeling really anything. And anyone who does computational modeling, whether they're into active inference or not, it it knows this and is going to have to confront that question, right? I mean, if we if we use uh linear regression to try and describe the relationship between X and Y, we're making an assumption of, you know, linearity is a valid way of describing the relationship between these things. And if you disagree with that then suddenly you disagree with the entirety of the model. Right? So so at some level you do still confront that fundamental question. And then ideally if you're working in a more empirical setting where you could fit it to data it's there that you know fingers crossed there's more validity to be found in your model beyond just you know hoping it works out. Um like actually Daniel has like some interesting like comments in the chat and like I want to take if you can like elaborate more on that like so I'm thinking in the perspective of like since active inference makes claims about what cognitive agents are doing like so you're coming at it like using active inference to model cognitive phenomena. I'm just thinking it makes normative claims of what cognitive agents themselves are doing. So I was curious about whether it can describe how cognitive agents themselves decide in which state space to use or maybe those are questions beyond that framework. This this is one of the kind of core um strange attractors is like map territory. I'll put some more map territory related discussions but I take a very strongly instrumental take like active inference the framework can't say anything about cars or planes or birds because it's a human created construct which can be applied by cognitive agents to make claims about systems. So the idea that the framework sort of like generically or unilaterally or already makes an opinionated claim about what a system it's like saying what what do linear regressions say about the relationship between this and that. It's like surely we don't want linear regression framework to be opinionated on such a kind of last mile problem and it's not. But you could use linear regression in a specific investigation to have datadriven claims about how this and that are associated. So again open to everyone's take obviously but active inference does not make claims period. It doesn't speak. People use active inference and hide behind apparatuses of theories to make it sound like well science says this or reinforcement learning says this would happen. It's like instead of just saying I am using reinforcement learning to say this happens. So that's kind of a grammatical thing, but actually it becomes really important because it's like is someone trying to bring the hammer of a whole framework down or like act like the framework is already giving opinionated stances and then it's like well then you can model different possibilities like if you say I'm modeling choice behavior in this mouse do these data better fit with a two or three state discrete model and then you're having a datadriven discussion about like what is the relative evidence for a two-choice model or three- choice model and how much does the increase in fit merit being penalized by adding more parameters. That's like the basian information criterion. So there are a lot of really nuanced ways to pursue structure learning and inference but none of them involve the framework speaking for itself somehow. And the first step in seeing that is just realizing that generative models are constructed artifacts. But it's not like well what is what does active inference say about dyslexia is is is essentially a category error question. Let me maybe rephrase myself there I guess. Um so like if you just want to say instrumentally like okay uh active inference instrumentally tells us how we should be designing our generative models about and how we want to capture model how we want to model cognitive phenomena and so if we want to model like we do structural learning right we make abstractions and if I want to model that phenomenon itself is this the right tool to use or like it offers us nothing on that perspective. Do you get my point? Like maybe that might be a more instrumental question. And so I'm just curious guy going into this because like I'm uh I coming on onto this idea of like uh if I want to let's say automate science itself like intriguing there's like a whole group like like this machine learning for science right like uh how do you make machine learning and techniques of doing science itself and you know it's right now a crazy idea in my mind but like I've been thinking about you know how humans really make abstractions and you know deal with like reality and break it down and so I'm like thinking okay like we do this cognitive phenomena of science like how do you capture that how do you model that itself and so that's why I was like thinking does this whole active inference framework tells us something about modeling or capturing that phenomena and maybe that's more of an instrumental question and yeah I might be asking questions that are not really relevant to chapter one but just curious about what you guys think about it >> if all you have is reward you're always going to need to to make a reward function somehow cover information gain. So if you want to model epistemic ecosystems that have goal seeking and open-endedness, then it's a natural framework. And I'll put some resources about like some of those like kind of science meta-cience active inference approaches, but a lot of them are are not executable code. >> Okay. that would be still useful for me to look at. >> I have a couple of more basic questions, but I didn't want to uh derail your presentation here, Andrew. I don't know if you should we save questions until the end. >> No, no worries at all. I'll um I'm just kind of like gathering together the the thoughts because the whole thing is that chapter one is always interesting because it's like oh this is interesting but also like we're quickly introduced to a lot of terms that won't be expanded upon till later in the textbook. So I'm not incredibly surprised that like here's me pulling up like an equation that appears four chapters from now and a second ago we had the kind of meta discussion of of choosing how to construct your model. I'm like, "Oh, yeah. Chapter nine has a good explanation of a of sort of these particular author's take on on, you know, there's there's constructing a model and then there's like the the fact that we are the mo as the modelers have our own understanding of that model. Like whenever we're doing the the model fitting process where we have data, we're trying to put together a model and we're making our own inferences about what the model should be to represent the the supposed model that the the the mouse in the T-maze is equipped with. Uh you know, so it's it's it just you know, infinite regress sometimes for for modelers. But um yeah like I I'll say as a as a basis for this chapter we are told that the the way the framework is set up is to uh assume that that it's interesting because it takes different kinds of sort of agnostic approaches to some things. uh for example, generative models, right? Like this book does not make a strong stance on saying that you that that you know a person has an explicit generative model or if it's such that the inference processes occurring in the brain are as if it has a generative model, right? Like there's no magical idea that we can go into the brain and pull out like this is exactly what the state space looks like, etc., etc. um as far as the sort of optimization function goes and free energy minimization it's there that it's it's really mainly there that that Fristen and um a great researcher Isura and some others are doing the empirical work of like actually you know working on validating the free energy principle and then from there it's like well if if if that's a kind of process or process theory that that we're willing to at least entertain then it's from there that we can start to entertain these models that utilize that kind of um optimization function. So um so we know from this chapter that we're kind of looking at a basian framework and in simple terms which it'll get more into it in chapter 2 but for those unfamiliar with that uh the general idea of like you you you have beliefs you receive new information you might use that information to update your beliefs right so like uh I think that this coin in my hand is a fair coin probability ility.5 heads,.5 tails. Um, and so I might flip the coin 100 times, realize like, oh, isn't it bizarre that 99 times out of a 100 it landed on heads, and the one time it landed on tails, I kind of like moved kind of funny and and it hit the wall or something and it just happened to fall on tails. I suddenly infer like, oh, I don't actually think anymore that this is a fair coin, right? It almost always lands on heads, right? So from there I might update my beliefs. This is not a fair coin. So I received new and sensory information from observing the coin in order to make a new you know inference. Like um the idea of this coin being a fair coin or a biased coin is already a way of splicing things out saying oh those are two discrete possibilities for this coin. Another way of looking at it is I might even try and estimate based upon those numbers of values what the bias of the coin is. So now we're in a slightly more continuous space, right? Like is the is the bias point, you know, instead of it being 0.5, it's like 0.99.01. That starts to look a little bit more like a frequentist uh paradigm if I break it down to that level. But nonetheless, receive information, update your beliefs. We're given some other uh terms. I've already mentioned perception and action. Uh so we have learning which is is similar to inference but it would be actually adjusting model parameters. Um that means like my beliefs about this coin it's 0.5 heads.5 tails. Um but I might over time adjust my belief my the very thing that I use to make inferences about this coin. I decide oh this is a biased coin. Now whenever I flip it I'm not going to expect 0.5 versus.5. I'm going to expect 0.99 versus 01 dramatically different. If I hold on to that, that's a new categorical probability distribution that informs me of what to expect from this coin and I I change that. That means I've changed my generative model. I've changed the parame outcome. Right? So to learn over time in more simple terms is to actually adapt to changing circumstances. not just react to them with the same model but actually changing the model itself the parameters within it. There's another aspect called structure learning which would be actually an actual change in the model like you add a new SPL a new slot in the categorical probability distribution or you remove one uh you know using some kind of information criteria. Um so so those are different notions in there. Chapter for those who are most interested in kind of the neuroscientific aspects of this neurobiological it's chapter five where you'll see a lot about message passing which is some of the most I mean those are the densest pieces of the maths here. Um but it it's there that we finally get discussions of like superficial versus deep pyramidal cells um interconnectivity between different neuronal populations. The idea of them passing messages between one another where you tend to have these kinds of hierarchal processes where there are these down like a downward past predictions coming from the brain and and uh uh rising uh prediction errors. they're actually being matched against incoming sensory observation. So it's like you have your beliefs about the world but you through sensory observation like start to realize there's contra uh contradictory information right and so it's from there that we start to update our beliefs like it it gets much more into sort of the the the more concrete like roots of of where this comes from and a lot of the work that that uh that that Friston's been doing the past several decades uh at this point with working on neuroiming data and then trying to formalize that into a scheme that could then be represented in a single chapter in a textbook. So it's quite dense. Uh but yeah, so um there there there's a lot more to this textbook and that we we frequently have folks who who join who are not exactly like excited to learn the maths but are more interested in the conceptual aspects. So I'm just trying to like get some of these words out here. I I know that I'm getting a little dense with with some of them because I kind of am trying to make my own in inferences about the interests of this group. But um yeah, it's uh I I do want to give a second to voice for anyone who hasn't had the opportunity to speak yet. Uh like if they have any kind of questions or anything they got out of this and and or or or if they want to chat about about something because it would be informative for me going forward of maybe what kind of things we could focus on for next time. Um, yeah, if anyone has any thoughts, I have a question for you. um regarding how is this um you know both free energy principle and active inference has been used in studying causal inference or say something like potential outcomes framework and things like that. >> Has there been studies or has there been papers that you can suggest that have looked into cause and effect and use of active inference? In other words, how do organisms can alter their environment and get and act on their immediate environment in in in in terms of uh causality? >> That's a good I mean because I I have my own personal interests in causal. I mean it this is and I I'm maybe it's just given the amount of time we ha we have left but so this might not be a satisfactory answer to that question because I don't I don't have any papers immediately that like are explicitly focused on active inference and causal inference but I will say uh as far as the general idea of doing something like so from my background we we do agent based modeling is a kind of computational paradigm which is you know but but this is done in in cancer research. You know this is done in in many different areas. And so the idea of like simulating computationally modeling some kind of entity or multiple entities within an environment where they uh you know may interact with that environment. they receive inputs, they release outputs, they receive obser some kind of observations, they they elicit some kind of actions back out into the environment. Um if if if we want to just take a kind of a for now like a softer approach to looking at causal inference and saying like oh well if if we track all of this over time by running it playing it out in terms of a simulation >> to where we can track which which which internal variables matter in these models that are leading to particular of emergent behaviors >> um and then looking at how those emergent behaviors then impact the environment. So, so there's a lot of kind of like um near simultaneity uh in looking at these things as kind of like an action perception loop. Uh when you model them, you do play them out over time. So you could, you know, in a toy example with a mouse and a teammates, you could run that for 60 trials, three time steps per trial in a discrete time setting. Um but so so you see that the idea is to model dynamics that play out over time and so necessarily there would have to be some kind of causality there. Um like in the sense that you know it's it's by the agent doing this particular action at this particular time that leads to this particular outcome in the environment which then leads to you know some kind of response from the environment. So um so so all in all I would say yes like the the the quick that the the shortest answer is like yes like a lot of people are trying to use these kinds of models like almost expressly to understand the dynamics and that would entail some kind of like causality. Um I mean there there are those who like include things like stochasticity and assumptions that you know what what happens whenever you're in environment that doesn't have like onetoone direct like deterministic rules. >> Um and then they try to see how agents might adapt to that kind of stochcasticity. Um or or maybe like they the the agents adapt to the amount that is deterministic but never becoming so precise that the the agent assumes it's in a a perfectly uh deterministic um environment if that makes sense. So, so yeah, there >> it does. And I think when you were speaking um Daniel um suggested a paper published in nature about the dynamic causal modeling of uh COVID 19 and I think that was a very good um you know it's a good example almost like a you know telltale example of what you were talking here in this particular situation. The other thing that interests me though is if you think in terms of the potential outcomes framework that what if yeah X were not to happen um you know what would be the change in the um in the state of Y. So if assuming that say X leads to Y then we say not X would be not Y right? So that's a causal premise. So the probability of Y given X yeah is what we get to see right. So if the if the X states were going to be changed Y would not occur. It seems to me that when we talk in terms of active inference, that underlying fundamental causal thinking still goes on in how an organism or how an entity kind of determines its position with respect to the sensory or you know otherwise environmental cues that that that organism is getting or that agent is receiving here. So my question was around that and um you know you've touched on what what we we were basically meaning. >> Yeah. >> It's just I was wondering if somebody has done a formal mathematical modeling of the potential outcomes framework using the active inference >> theory. >> Yeah. No, absolutely. I mean I I think that every model that I've seen thus far that expressly is like an active inference model and that's the the framework and approach that people are taking usually what they're doing when they construct these models is to not just construct them in one way and and say deterministically like this is how this works but they will play around with the parameters and run them with different kinds of benchmarks not just for the sake of like oh I'm trying to make you know an extremely amazing predictive model but rather like if they're trying to simulate empirical phenomena then yeah that's exactly what they're trying to do like entertaining counterfactuals what would happen if things went this way instead or we had these kind of parameters at the start or these conditions or otherwise I'm personally very interested in things like that I've written a um I presented at a University of Pennsylvania with computational social scientists I gave like a three-hour talk on like multi- aent simulations where you have agents learning with one another and say like if you have these this kind of network structure between the agents, then we tend to see these kinds of emerging behaviors as opposed to these with these kinds of network structures and and so so yeah, I mean it's it's those kinds of questions would definitely be interesting to a lot of folks who are in active inference and vice versa. >> On on that note, Andrew, um would you be kindly u able to share that uh copy of the of that uh presentation that you gave if you have one? uh if you have a copy of that as well. I'll tell you why I'm asking you this is because at the moment I've got a PhD student who's looking at um systems thinking and uh you know systems modeling of um people accessing uh health services uh of those people who have got um you know postacute covid syndrome and how what kind of um you know health service organizations are working and she's taking on a systems thinking approach to it and when I was listening to your um you know conversation I was wondering that this kind of thinking would really benefit um the student to look into these you know uh I mean some of the things that you're talking about and we can have a discussion around that so if you can kindly you know provide that be excellent >> no absolutely um um I'd be happy to do that I I put into the chat this is uh it's a bit messy but but one of my GitHub repositories that has kind of a collection of just a couple of the talks I've given but also O has some code and more importantly it has this this slideshow presentation that is kind of a intro to active inference and it has various links. Um, I'll be honest, I've been so busy over the past year that like the the the work I've done deserves to just be a paper and I should work on publishing, but I've had no time. Uh, unfortunately, so it's not the cleanest, but but I would share that that link with with them and and um um let them have a look if there's anything of of interest. Yeah, for sure. Um because that's what I was trying to teach. I mean with with computational social science we're already doing the agent-based modeling active influence would not be a dramatic revolution rather it would just be aligning it much more thoroughly with work that's being done uh in in decisionm and inoscience that goes beyond some of the kind of minimalist assumptions that are made about human behavior. Uh it's sort of like instead of thinking of like profit maximization for an economist like maybe think of free energy minimization like we do more seeking a monetary value. Yes. >> Uh right. And and yet in terms of like computational tractability you know not every student who does these things has an entire data center available to themselves. Well this is why there's been a lot of simplicity in agentbased modeling for so many years right. It's it's a it's a empirical problem of us not having the the technology that we need to really run some kind of massive large scale computation. And that's something that's very attractive about the free energy principle, right? Like in so far as like it's been uh empirically validated at at particular levels in the particular experiments that that have been carried out to to test for that. Um like it it it's still a tractable problem. It's it's you're still ultimately looking at one major quantity or perhaps two if you're looking at planning and expected energy, but still not not too much more than that. Uh right. So so replace the dollar sign with something that is is is much more thoroughly kind of, you know, thought through and and and kind of bears in in other areas of literature, right? beyond just the silo of just you know economics or something like that. >> Exactly. And that's I think is the biggest attractiveness of uh of of of an approach here. Anyway, I'll let uh you know John continue with his uh observations and thank you everyone because while we were having this conversation uh Abas and uh you know um and Daniel both of you u put up some excellent resources that I'm looking at and you help John >> um and briefly because we we are unfortunately out of time that kind of flew by in some respects but I was Just looking at uh Rachel's comments uh applying these principles to computation modeling your own models model revision computational psychiatry. Yeah, cool. >> We have a lot of overlap I think and I figured it would be easier for me to just share a link uh versus trying to explain. >> Yeah. No, cool. It's uh neurocomputational approaches and neuro neuropenomenology. That's I'm currently working with I'm working with a team where it's it's very expressly like kind of like a working with a team includes a clinical therapist as well as someone who kind of does like behavioral research uh regarding like avoidance and uh and so we've been working on this problem uh this project for for trying to model uh certain dynamics uh that they're commonly found in the symptomat PTSD. Um, and so >> I saw that listed in your projects. I would be interested in hearing more. Um, but yeah, in general, anything that you want to link back to your work with neuro imaging or general uh behavioral studies would be interesting. >> Yeah.

No, that that's great. And there's a lot of so there's been a lot of literature on these on these things. Um what one issue I frequently find is that people are repeatedly trying to do discrete state spaceon uh kind of theoretical model. >> Um I like Ryan Smith's work that I that I referenced earlier in the chat um because he kind of covers all the grounds, right? So and he you know he has a lab and so he's carried out experiments where he fits these active inference models to to empirical data. So, I would definitely like kind of point to him as a quick reference. >> Yeah, that that's like my main question at this point is like is anyone actually using this or are we just are we still just talking about the possibility? And it seems like maybe we're at that turning point. Um >> yeah, >> but yeah, that's vibes >> for sure. And it's uh there's a good it it has a funny title just because it's like slightly funny, but it it's called Gut Inference. Uh and I'll share that. Um, and that is a study that uh Ryan's been playing out for some time and some of his studies he'll like repeat them or he'll do them with the same participants a whole year later. So I just I appreciate the depth to which he goes to like focus on the actual model fitting and and validation. But with this paper, it's just like um this is kind of studying reaction times and things like that, but still it's it's drawing from multiple forms of data. And um I just I think it's without going into a lot of detail, I think it's a nice example of some of the work he's done. Um yeah and then and then sticking to the textbook um a lot of the predictive if you see the phrase predictive coding or if you look at like chapter 8 where it talks more about continuous models those are the things that will more directly be applied to neuroiming not just neuroiming data but but given like the the speed at which it's collected is continuous data like those are probably sort of the exemplars for how to get started on thinking of of how to apply this to to that data. So, um, >> got it. Thanks a lot. >> Yeah, for sure. Um, and then I think we're just about out of time, but John, did you did you want to throw something out there? I said a very basic um terminology question and is I assume this will come up again later but um is it's the first time when he's introducing or the book is introducing um kind of the what B optimal you know active inference requires they it talks about minimizing highs and then later on the chapter as talking about the if the brain's doing anything it must be minimizing free energy >> um >> and I'm I'm kind of noticing those like I don't think it's free energyy's technically been defined yet. Um I think it was loosely listed as like a proxy for surprise. Is that is that an appropriate way of understanding what's are those terms interchangeable in the sense that free energy is a is a is a you know approximate measure of surprise? That just seems like a very core uh piece of what's going on here. >> No, for sure. That's a that's a very good question and uh I'm I'm like looking for it right now but somewhere later in the book they actually like provide this like simplified table. Um page 50 I I won't go too much into it but for now like your intuition is correct. It is kind of like a proxy. However, they're definitely like not necessarily equal. Um if only they were. Uh so so surprise basian surprise the issue with trying to directly compute that as a quantity is that it's frequently intractable. Um it doesn't allow for like marginalization in all cases. And so what what happen essentially what's happening is that like you know if I want to go down the street to get a drink from a convenience store I don't necessarily have to think about the spatial layout of the opposite side of the world. Like I don't have to think necessarily about everything that I could possibly think about in order to optimize. Right? So that that's a crude way of saying that, but but it it's it's so free energy minimization involves like variational inference and it involves certain kinds of rules and and for marginalizing over probability like state spaces and so um free energy is a tractable proxy for basian surprise. And >> if you can minimize that quantity as much as possible, you would necessarily be minimizing surprise, you just might not be getting it down to um as low as possible. So free energy in this case is actually it's like an upper bound on surprise, right? So if you can bring that down and and then whenever you actually break open those equations that I was sharing on the screen earlier um like if you can minimize your KL divergence as much as possible to where it's virtually zero it's in that case that basine surprise would necessarily equate to free energy right so it's like you can get close to actually minimizing surprise itself but if your KL term is still greater than zero then you're still just looking at like a proxy. Um, I hope that wasn't too deep, but maybe that gets across. Yeah. >> And the surprise here is the surprise between the discrepancy between your preferred state of the world and your actual observations, not between your expected state of the world and and or pred not between your predictions and observations, but between your preferences or your goals and observations. That's at least that's you introduce it in uh you're on whatever page this is 1.4.1 4.1 the discrepancy between your preferred state which is seemed to me not how I would have traditionally thought of that but >> yeah no also a good point that you brought up and uh like for sake of time how to think of so with I mentioned earlier there's variational free energy and then there's expected free energy um actually with variational free energy that would involve just what you expect not necessarily what you want so to speak uh for those interested in like dynamic chemical systems and thinking about like attractors like the notion of an attractor. You can think of preferences in expected free energy like what what you want so to speak. Uh what you can think of them kind of like a reward distribution in a sense. It's like oh a point8 for the cheese in the versus like you know an aversive stimulus in the other arm of the teams would be a 2. like that you can look at that those as expectations but that are acting as attractors that are actually attracting the agent towards them. Um I don't know if I'm doing the best job in describing it but it's it's kind of like it's both. It's both of the things you said. It's variational free energy relate will relate to what you expect regardless of what you actually want in that instance. Then expected free energy is about how do you score what you should do in order to get to your preferences? Not just what you expect, but like your your preferences. They kind of get equated in that sense. So um with variational free energy, it's like that could that could change like what you like what you expect from the world around you could change over time through learning or otherwise. Meanwhile, my blood oxygen regulation in my body needs to be in a certain range otherwise my I'm going to dissipate, right? Like that would be an equivalent for something like preferences that you could think in like a hardline homeostatic seeking kind of way is like I need to have so much so much oxygen and glutamate and so on in my body otherwise I'm I'm done for. Right? So that it's through expected free energy that I figure out like, oh, I need to go eat some food in order to fulfill these preferences. So preferences aren't a flimsy like do I want chocolate or vanilla. They they like in thinking about them, you can also think about them as like how does a an organism stay alive and maintain its its well-being. So that that's the significance of preferences. Um, I would say just look a little bit more about uh the equations for variational free energy and expected free energy and you'll just you'll see a lot more of that just literally in the next chapter and then chapter four we'll kind of bring it back around too. So >> yeah, I I'm assuming we'll come back to that. I I'll just pin that because I don't want to I know we're over time, but I do want to that's one of the things I want to get ironed out to be clear. I still >> It's a good Yeah, it's a good question because it's like if you only ever expected what you wanted, right? Then it's like how would you actually be able to account for what actually tends to happen, right? Um you delusions, I guess. Uh so yeah, we we can get more into that though. But this was a good session. So thank you all again for for attending and uh look forward to hopefully seeing you next week. Uh next week it will uh well maybe not next week for you all for the evening session. The evening session next week we'll be doing chapter six for the other cohort. Again it's it gets a little confusing. Feel free to email uh Daniel or I or the the main uh institute email address. It's on the on the site. And uh yeah, thank you all so much. >> Two weeks from now, right, was when we'll do the next >> Okay. >> Exactly. And I'm sure this is >> I'm sure this is clarified in the documentation, but is there like a chat that we can all discuss ideas over the next two weeks asynchronously? >> Yeah. Yeah, for sure. >> And if that's in the docs, just tell me to go look at the docs. Happy to do that. Yeah, I think I think uh whenever uh I think it's Daniel who sends out the the emails for the the textbook group. Um he should have included any kind of like rel like significant links in there. Um two two quick things are like one there's a discord channel there all kinds of folks in there who are you know approaching different things. So you know uh engage with that as you like and then and then there are more general channels for discussing things but then there's also the textbook group on the KOD where you can kind of get into like more direct questions about the textbooks. >> Oh sweet awesome. Thanks so good to meet everyone. legs. So anyway,